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Faculty of Engineering

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Numerical Studies of Transition and Turbulence in a Flat-Plate Boundary Layer

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Master's Thesis, Bachelor's Thesis, Project Thesis

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Contents

List of Figures

List of Tables

Acronyms

TBL turbulent boundary layer.

ZPG zero-pressure gradient.

1 Introduction

Refer to equations using the following way. In equation (A.1) we can replace $G(x)$. As seen in Fig. A.1, however, the referred object always needs to come before the corresponding text. We can refer to literature **?** as an important citation (see for instance **?**, and others).
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$$G(x) = \sum_{0}^{N} \sin^2(x) \tag{1.1}$$

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1.1 Important stuff

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Figure 1.1: The LSTM logo with the latest colours.

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Table 1.1: A table has the caption above.

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2 Basics

2.1 DNS

2.2 Turbulent boundary layer theory

- boundary layer develops const in flow direction - with increasing boundary layer thickness - wall shear stress not know a priori - behaviour in inner layer ess. same as in channel flow - equation for pressure gradient - definition of boundary layer thickness, displ. thickness, momentum thickness - corresponding Reynolds numbers - shape factor

Turbulent boundary layers ..... boundary layers occur when fluids with non-zero viscosity stream over a wall that is not moving. IN these setups the so called *no-slip* boundary condition impacts the stream which states that the velocity of the fluid in wall-normal direction and in streamwise direction need to be zero directly at the wall Schlichting and Gersten (2017) since there is no relative movement between the wall and the fluid. The velocity far away from the wall, the so called *free-stream velocity*, is usually much higher than zero which implies the existence of a transition region where an acceleration of the fluid occurs. This so-called *boundary layer* is placed close to the wall. Depending on the Reynolds number the thickness of the boundary layer varies such that the boundary layer decreases with increasing Reynolds number Schlichting and Gersten (2017). The flow in the boundary layer can behave laminar or turbulent, whereas in this thesis we will focus on turbulent boundary layer (TBL). The boundary layer grows in streamwise direction and can be described via multiple characteristic quantities. The introduction of these quantities requires a consistent coordinate system: We define x as the streamwise direction with its corresponding velocity u, y represents the wall-normal direction with velocity component v and z describes the spanwise direction with velocity component w. In the free stream far away from the wall we assume U_0 being the dominant velocity component. For simplicity this free-stream velocity is often set to be 1. The velocity component in streamwise direction inside the boundary layer is named as $u(x, y)$. To describe the thickness of a boundary layer mainly three different parameters are used: the boundary-layer thickness $\delta_{99}(x)$, which describes the distance from the wall at which the velocity $u(x, y)$ derives 99% of the free-stream velocity; the displacement thickness δ^*, which defines the displacement of an inviscid fluid to preserve the same mass flow; and the momentum thickness θ which define the displacement of an inviscid fluid to fullfill the conservation of momentum. These quantities can be derived with the following equations

$$u(x, \delta_{99}(x)) = 0.99U_0 \tag{2.1}$$

$$\delta^*(x) = \int_0^\infty \left(1 - \frac{u(x,y)}{U_0}\right) \mathrm{d}y \qquad \theta(x) = \int_0^\infty \frac{u(x,y)}{U_0} \left(1 - \frac{u(x,y)}{U_0}\right) \mathrm{d}y \tag{2.2}$$

In this thesis we use the momentum thickness θ to characterize the height of the boundary layer, since it can be measured more accurate than the boundary-layer thickness. To describe the boundary-layer with the Reynolds number, different Reynolds numbers based on the introduced displacement measures are defined as

$$\mathrm{Re}_x = \frac{U_0 x}{\nu}, \quad \mathrm{Re}_{\delta_{99}} = \frac{U_0 \delta_{99}}{\nu}, \quad \mathrm{Re}_{\delta^*} = \frac{U_0 \delta^*}{\nu}, \quad \mathrm{Re}_\theta = \frac{U_0 \theta}{\nu}. \tag{2.3}$$
$$\nu : \text{viscosity}$$

From the RANS we can derive the boundary layer equations for a 2D boundary layer over a fixed flat wall.

$$\frac{1}{\rho}\frac{\mathrm{d}p_w}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}y}\left[\nu\frac{\mathrm{d}U}{\mathrm{d}y} - \langle u'v' \rangle\right] \tag{2.4}$$

With this equation we can define the wall shear stress τ_w as

$$\tau_w := \mu \left.\frac{\mathrm{d}U}{\mathrm{d}y}\right|_w = -h\frac{\mathrm{d}P_w}{\mathrm{d}x}. \tag{2.5}$$

To study the phenomenons in a boundary layer it is required to define so called viscous scales since the ranges of dimensions and velocity variations are much smaller than in the free-stream. Based in the wall shear stress the friction velocity u_τ, the viscous length scale δ_ν and the friction Reynolds number Re_τ are introduced as

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \qquad \delta_\nu = \frac{\nu}{u_\tau} \qquad Re_\tau = \frac{\delta_{99} u_\tau}{\nu} \tag{2.6}$$

With these quantities the original length and velocity scales can be nondimensionalized in wall units as (fundamentals of tubr flows)

$$y^+ = \frac{y}{\delta_v} = \frac{y u_\tau}{\nu}, \qquad u^+ = \frac{U}{u_\tau}. \tag{2.7}$$

The advantage of these non-dimensional scales is that they are universal for all wall buonded flows what enables the comparison between flows with different physical dimensions. XX unbedingt QUELLE XX
something about plus units

2.2.1 zero pressure gradient tbl

clauser parameter
why is it important?

2.3 Optimization??

2.4 Tools

XX das hier vlt in Basics?? XX
To work on this project multiple tools were used. Most of the fluid-flow simulations (korrekte bezeichnung??) were performed with DNS (see XX section 2.1). To perform these simulations

the spectral element code NEKO was used. Blabla über neko When running simulaitons with NEKO it is possible to take statistics of the simulated flow during run time. Multiple settings about how the statistics should be recorded can be defined in the case file of the simulation. For example it is possible to define one or two axis along which the statistics should be averaged. These statistics are used to extract and compute certain measures of the flow. To do this some postprocessing of the recorded statistics is necessary. This is done in PYTHON where the package PYSEMTOOLS is included. With this package data recorded in NEKO simulaitons can be extracted and interpolated from an SEM mesh into any arbitrary points (zu nah an github description). Then multiple statistical measures like the budget terms can be calculated. In addition to the DNS, some RANS simulations were performed as well. To run these simulations the open source software OpenFOAM is used. THis software is able to perform RANS simulations of fluid flows with diffenret turbulence models. To extract velocity fields etc from the simulation Python is used. All simulations are performed on the cluster of the RRZE of the FAU. Blabla über cluster

3 Methods

In the following chapter the methods used in this thesis are presented. In the beginning the used tools are introduced. Then the workflow of the handled project is described and in the end the used optimization method is discussed briefly.

3.1 Channel simulations/cooler name für 'Anfangssimulationen'

3.2 plane TBL simulaitons

3.3 curved TBL simulations

- meshing - imposing bc - case settings
In the next step to derive the curved ZPG TBL we first need to create a general setup for the curved TBL. As a basis we use the setup of the plane TBL and add a curved part at the end of the domain of the plane TBL. The plane part is necessary such that the TBL can develop properly since we want $Re_\theta \approx 1000$ when the curvature starts. The curved part has a constant radius and develops in negative wall-normal direction. Therefore, the TBL experiences a convex curvature. At the end of the curvature another straight domain part is added which is called *straight outlet*. There is a smooth transition between the curved part and the straight outlet because it follows the tangential direction of the end of the curved part. The complete set up is shown in fig XX. The straight outlet is applied to reduce the impact of the boundary conditions at the outflow. The pressure BC as the outflow forces the pressure to be zero. Since the pressure does XX in the curvature, this outflow condition would impact its behaviour and therefore the development of the TBL. To reduce this impact the straight outlet is added. XX general dimension of domain XX The mesh is generated similar to the one used for the plane TBL and adapted to the shape of the domain. In the straight outlet the mesh is coarsened since this part of the domain is not relevant for the study of the curved TBL.

The imposed boundary and initial conditions are the same used for the plane TBL. However, they need some modifications due to the curvature. The boundary conditions are summarized in the following table: XXXXXXXX As initial condition the blasius boundary layer was used similar as for the plane TBL. In the curved part the blasius profile is rotated such that it is formed in wall-normal direction.
This setup was used to created different variations of curved TBL. We tested two different radii such that we can study the impact of the amplitude of the curvature. To do this we the radii of $R_1 = 100 * \delta_{99}$ and $R_2 = 500 * \delta_{99}$ with δ_{99} measured in the plane TBL at $Re_\theta = 1000$. For both radii we tested the angles of 30° and 90° of a circular arc.
XX Tabelle mit Fällen XX

3.4 ZPG Generation

In the next step the procedure to generate a ZPG in the curved TBL domain is introduced. This procedure is done on OpeFOAM using the Bayesion optmizer code (ref: Eman etc XX, ref: basics XX). As described in XX the code is able to optimize the shape of the domain to achieve a given pressure gradient function, which in this case is ZPG. To do this the simulaiton setup has to be transferred to OpenFOAM. In the following paragraph the necessary steps are described in detail.

3.4.1 OpenFOAM setup

General settings RNAS model

Mesh Generation The transfer of the curved TBL simulation setup to OpenFOAM includes some adaptations which will be presented here. First, the mesh has to be regenerated. It is not possible to use a mesh created in GMSH and transfer it to OpenFOAM, since the code has to adapt the domain and the mesh after every iteration of the optimization. Therefore, the domain has to be regenerated in every iteration with `blockMesh`. As a consequence the domain has to be defined in `blockMeshDict`. The plane part is meshed with rectangular blocks. The domain close to the lower wall has a progression factor included such that the resolution increases towards the wall. In the curved part the lower wall is defined through a circle arc similar to the GMSH mesh used in NEKO. The upper wall is determined by a spline. The points defining the spline are the coordinates x_{θ_i}, y_{θ_i} of the *design parameters* θ_i whose position is modified to minimize the *objective function*. The optimization code is constructed such that it modifies position of the *design parameters* in wall-normal direction. Since the *design space* is curved, the wall-normal direction does not correspond to a fixed axis. Hence, their position has to be transformed in $x-$ and $y-$ coordinates to define the spline correctly in space. Compared to the domain used for the DNS, the height of the domain is increased to $\delta_{99} * XX = XX$. This modification brings some simplifications when the results of the optimized OpenFOAM simulation are transferred to the DNS domain which will be described later (see XX).
XX HIER BILD VON MESH XX

Initial conditions or boundary condition ??XX The quantities that need predefined initial conditions depend on the chosen RANS model (XXQuelle). Since in this project the $k-\omega-sst$-model is used, it is necessary to define initial conditions for turbulent kinetic energy k, dissipation ω and velocity U. This is done due to the imposition of an initial field at the inflow of the domain which is extracted from the previous DNS simulation of the curved TBL (see XX). In the postprocessing of the DNS simulaiton the streamwise position in the plane part of the domain where $Re_{theta} = 1000$ is reached is identified. At this streamwise position the fields of U, k and ω are extracted along the height of the domain. In order to do this the postprocessing of the DNS simulation data is repeated for the plane at $Re_{theta} = 1000$. With the PYTHON package *fluidfoam* it is possible to read an OpenFOAM mesh. This is used to extract the points of the OpenFOAM mesh at inflow. These points are used as interpolation points to perform the postprocessing. As described earlier the domain used in OpenFOAM is higher than the one for the DNS simulations. The usage of this higher domain would lead to interpolation conflicts since there are no data for the upper interpolation points. To avoid these conflicts the interpolation is performed on an OpenFOAM domain which has the same height as the neko domain. The values for U can be extracted directly, whereas k and ω need to be computed. To use these fields in the higher OpenFOAM domain additional values for the upper part of the inflow are manually added. For this part of the inflow the last interpolated value is copied and added in the remaining cells.

3.4.2 Optimization settings

- how many design varibales - where are they located - why? how many iterations - all settings done for the optimization

The results of this simulation are used as inflow conditino for the following RANS simulations in OpenFOAM. More precisely the measure of U, k and omega at the positoin of $Re_theta = 1000$ are extracted. Then the OpenFOAM simulation is set up such that the profile at $Re_theta = 1000$ from the DNS is used as boundary condition at the inflow of the domain in the RANS simulation. XX hier noch das OPenfoma domain höher ist XX With this simulation setup the optimization process is started. The first step of the optimization loop is to ste new boundary points q. In the beginning (if there are less then XX in GPinit) the points are taken randomly in the given parameter range. in the next step. The domiain is build with the new boundary points and the OpenFOAM simulation is started. When the simulation is finished, the postprocessing is performed to compute the current Clauser parameter β and the residual R. As a last step the files and XX are updated. If the optimization was performed for a certain number of steps the new boundary points q are selected with the bayesian optmization. <the optimization runs until ???XX, sich that the ZPG is ensured. In the next step the velocity profile for the optimized domain is extracted. The values are extracted from the RANS simulation at the wand normal position which corresponds to the topwall of the DNS domain. This velocity profile is imposed to the DNS setup and then the DNS simulation is performed.

3.4.3 data extraction

- how to extract data from openfoam and impose them on neko domain?

3.5 simulations with ZPG

3.6 General workflow

In this section the general workflow followed during this project is presented.

XX Hier noch allgemeines Vorgehesn beschreiben sonst zu detailiert im folgenden XX

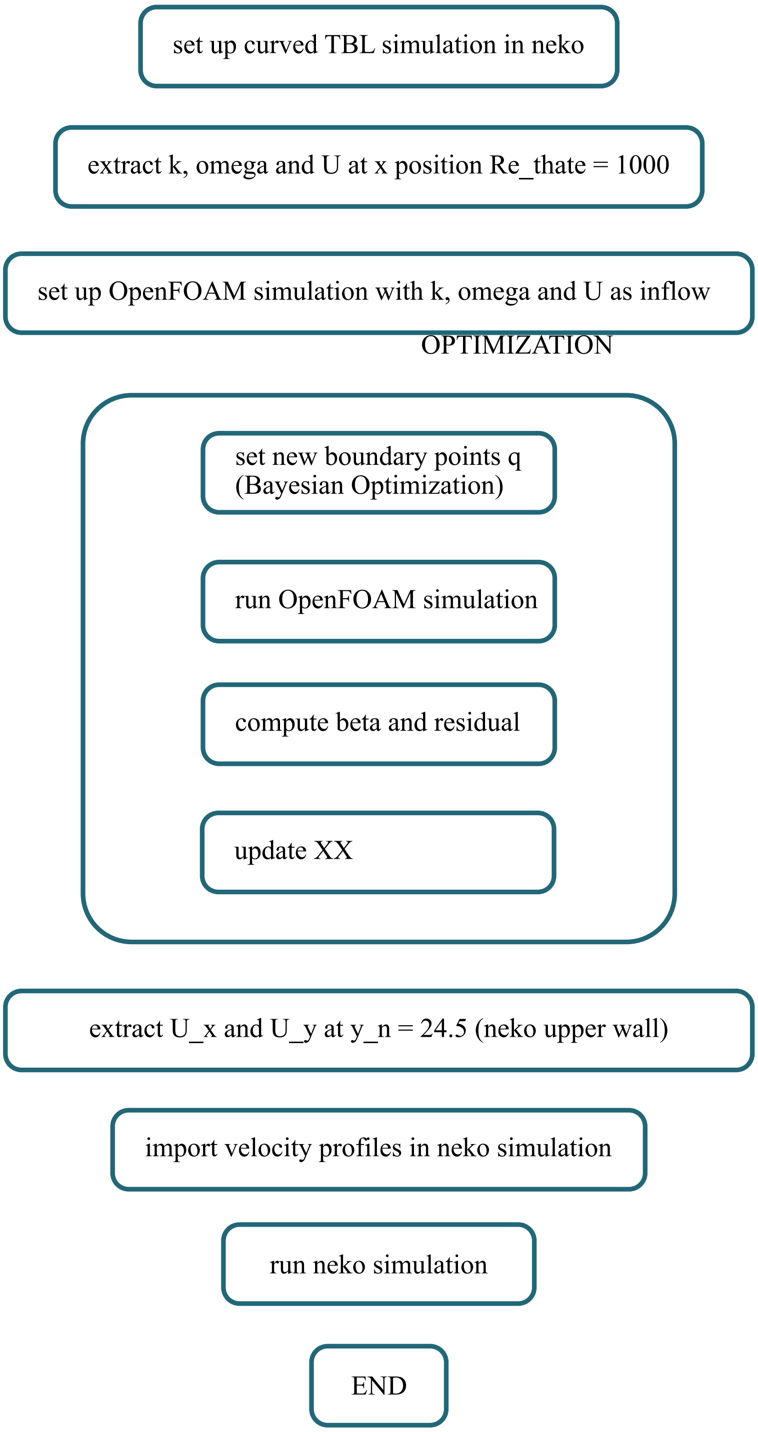

Figure 3.1: Worklow

To get familiar with the NEKO software an example case of the neko example is simulated. We decided to use the channel case since there are some similarities compared to the TBL (see chapter 2). The obtained simulation results as well as the postprocessing procedure are verified and then validated with reference data. In the following step the set up is switched inot a TBL. To do this a new domain with a new mesh is created. At this point a plane TBL is analysed first. A detailed description of the setup can be found in XX. This setup is validated with reference data of other plane TBL simulations eg XX XX. As with the cahnnel simulations, these simulations are used to validate the simulation setup and the postprocessing procedure. IN the final part of the project the curved TBL is analysed. Therefore it is necessary to build a simulation setup where a ZPG is ensured such that only curvature effects can be analysed. To create such a setup multiple steps are necessary which are depicted in Figure 3.1. First a working setup of a curved TBL without pressure modification is established in NEKO including the domain size, case settings, vector rotations etc. which are descriped in detail at XX. With this setup a simulation is performed.

- einarbeiten in neko software –> channel simulations - perform simulations and verify results with reference data
- create TBL domain - perform plane tbl isimulations and validate the results with reference data
- create curved domain - set uo neko settings for working simulaiton - copy setup in OpenFOAM
- run optimization
- beta (clauser parameter) at the lower wall as target measures - odjective function: distnace between beta at the bewertungspoints q nad prescribed beta along the lower wall (at the same x position)
- extract results (U, ??) from OpenFOAM - impose these data as boundary condition at top wall in neko simulation - perform neko simulation
- do this for different domains: why these domains?

Bibliography

H. Schlichting and K. Gersten. *Boundary-Layer Theory*. Springer Berlin Heidelberg, Berlin, Heidelberg, 2017. ISBN 978-3-662-52917-1 978-3-662-52919-5. doi: 10.1007/978-3-662-52919-5. URL `http://link.springer.com/10.1007/978-3-662-52919-5`.

A Einleitung

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Figure A.1: The LSTM logo with the latest colours.

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